

Another Way to Model LGD

Probabilistic Regression via Quantile-Based Classification

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1 Introduction

Loss Given Default (LGD) is a foundational concept in credit risk modeling. LGD measures the credit loss a bank is likely to incur due to an obligor default [2], and is widely used for regulatory capital calculation, pricing, and management reporting.

In statistical modeling, LGD is defined as follows:

$$LGD = \frac{EAD - \sum_{t=1}^n \frac{R_t}{(1+r)^t}}{EAD} \quad (1)$$

Where:

- EAD = Exposure at Default (total outstanding balance at default)
- R_t = Recovery cash flow received at time t
- C_t = Workout/collection costs incurred at time t
- r = Discount rate (effective interest rate or cost of funds)
- t = Time period index ($t = 1, 2, \dots, n$)
- n = Final recovery period (time of last cash flow)
- $\sum_{t=1}^n$ = Summation over all time periods from 1 to n

LGD has a non-standard distribution, usually bimodal, with concentration around 0 (no credit loss) and 1 (full credit loss) as Figure 1 illustrates based on the dataset from [1].

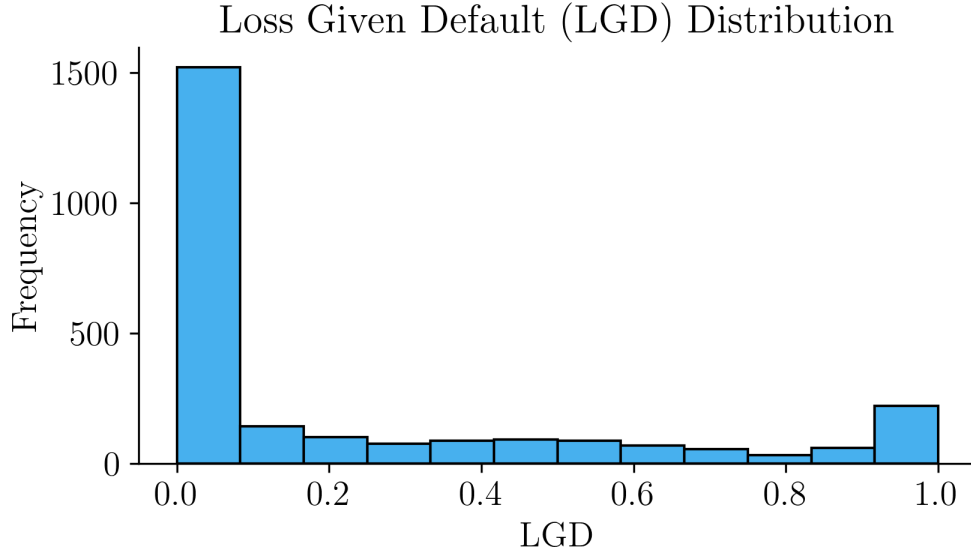


Figure 1: Loss Given Default (LGD) Distribution

This particular aspect of the LGD distribution motivates several alternative methodologies for statistical modeling of this risk parameter.

2 LGD Modeling

In this section we cover conventional approaches to statistical modeling of LGD¹ and offer an alternative methodology focused on modern machine learning applications.

2.1 Conventional Approaches

The most common approach to modeling LGD is a linear regression model. It relies on the assumption of normality of errors, which is typically violated. However, due to its simplicity and interpretability of results, it is often the go-to method in many situations.

Alternative methodologies include modeling LGD in two steps:

- Estimating the probability of loss: $P(LGD > 0)$
- Modeling the expected loss given default: $E[LGD \mid LGD > 0, X]$

¹Note that depending on model design and modularity, LGD estimates can further incorporate cure events, debt sales, and other outcomes within a single modeling framework, though such extensions are beyond the scope of this paper.

The resulting LGD estimate is then obtained by combining these components:

$$E[LGD \mid X] = P(LGD > 0 \mid X) \cdot E[LGD \mid LGD > 0, X] \quad (2)$$

While this approach better captures the bimodal nature of LGD and accounts for zero-loss events, it introduces additional complexity: two separate models must be estimated and validated, each requiring its own set of assumptions and diagnostics.

2.2 Probabilistic Regression via Quantile-Based Classification

Standard regression assumes constant variance. However, LGD exhibits distributional complexities:

- Some borrowers display low-or-high LGD patterns, leading to bimodality
- Different tail behaviors by segment result in skewness

To account for this, we can reformulate the problem as multiclass classification over discretized LGD bins, following the approach outlined in [3].

2.2.1 Methodology

Rather than predicting a point estimate $\hat{y} = E[LGD \mid X]$, this method predicts the full conditional distribution $P(LGD \in \text{bin}_k \mid X)$ for $k = 1, \dots, K$ bins:

- Discretize the LGD variable into K bins using quantile-based binning
- Train a multiclass classifier to predict $P(LGD \in \text{bin}_k \mid X)$
- Obtain probability distribution of LGD: $\mathbf{p}(X) = [p_1(X), p_2(X), \dots, p_K(X)]$
- Compute LGD point estimate: $\hat{y} = \sum_{k=1}^K p_k(X) \cdot v_k$

Here v_k is the midpoint of the k -th bin. Unlike standard regression with homoscedastic assumptions, this approach naturally models the characteristic bimodal LGD distribution with mass at 0 and 1.

2.2.2 Calculation Example

For example, let's consider 8 quantile bins ($K = 8$), whose midpoint values v_k are shown in Table 1.

k	q_1	q_2	q_3	q_4	q_5	q_6	q_7	q_8
v_k	0.0007	0.0065	0.0209	0.0558	0.1790	0.3989	0.6982	0.9375

Table 1: Bin values v_k (8 bins).

Table 2 shows five sample rows of multiclass probabilities $\mathbf{p}(X) = P(\text{bin}_k \mid X)$ for $k = 1, \dots, K$.

ID	q_1	q_2	q_3	q_4	q_5	q_6	q_7	q_8
1	0.095	0.058	0.066	0.075	0.162	0.163	0.151	0.229
2	0.326	0.119	0.126	0.112	0.082	0.096	0.076	0.063
3	0.312	0.099	0.108	0.115	0.096	0.116	0.081	0.074
4	0.412	0.132	0.124	0.114	0.078	0.048	0.052	0.041
5	0.097	0.111	0.035	0.043	0.060	0.234	0.248	0.172

Table 2: Five sample rows of multiclass probabilities $\mathbf{p}(X)$.

The LGD point estimate is then computed as:

$$\hat{y} = \sum_{k=1}^K p_k(X) \cdot v_k \quad (3)$$

where v_k is the bin value.

For row 1 in Table 2, the LGD point estimate is:

$$\begin{aligned} \hat{y} &= 0.095 \cdot 0.0007 + 0.058 \cdot 0.0065 + 0.066 \cdot 0.0209 + 0.075 \cdot 0.0558 \\ &\quad + 0.162 \cdot 0.1790 + 0.163 \cdot 0.3989 + 0.151 \cdot 0.6982 + 0.229 \cdot 0.9375 = 0.421 \end{aligned}$$

2.2.3 Applications

Below we outline some of the advantages of this approach in practical risk management applications.

Regulatory capital calculations

Direct extraction of any percentile for stress testing and expected shortfall:

$$\text{LGD}_{99} = F^{-1}(0.99 \mid X), \quad \text{ES} = E[\text{LGD} \mid \text{LGD} > \text{VaR}, X] \quad (4)$$

Stress testing

Probability mass can be shifted toward higher LGD bins under stress scenarios without retraining.

$$p_k^{\text{stressed}}(X) = \begin{cases} 1.5 \cdot p_k(X) & k \in \{\text{top 3 bins}\} \\ p_k(X) & \text{otherwise} \end{cases} \quad (5)$$

Followed by softmax normalization:

$$p_k^{\text{stressed}}(X) \leftarrow \frac{p_k^{\text{stressed}}(X)}{\sum_j p_j^{\text{stressed}}(X)}.$$

Economic capital and IFRS 9

The model supports both point-in-time (PIT) and through-the-cycle (TTC) estimates:

- Downturn LGD: 90th percentile for economic capital under Basel requirements
- Scenario-based ECL: Reweight probability distributions across macroeconomic scenarios for IFRS 9 lifetime expected credit loss calculations

By maintaining the full conditional distribution rather than point estimates, this approach enables flexible adaptation to diverse regulatory and business use cases from a single model.

2.2.4 Experiments

In our experiments, we use the LGD dataset provided in [1] to model LGD as a function of Loan-to-Value (LTV) and loan purpose. The dataset is split into 70% training and 30% test sets. For the quantile multiclass model, we use 10 quantile bins for the LGD target.

We compare the proposed method against a baseline CatBoost regressor on the original LGD and log-transformed LGD distribution:

- Quantile-based (midpoint): Bin representatives are the midpoints of bin edges
- Baseline regressor: Standard CatBoost regression with squared loss

We evaluate model performance using Somers’ D (D_{xy}), a standard rank-ordering measure for LGD models, and Root Mean Squared Error ($RMSE$), which quantifies prediction accuracy.

In Table 3 we report the results of experiments on the test set.

Model	Dxy	RMSE	Dxy (log)	RMSE (log)
Quantile Multiclass CatBoost	0.3185	0.3299	0.3171	0.3100
CatBoost Regression	0.2885	0.2905	0.3523	0.3877

Table 3: Performance summary

Experimental results indicate that the quantile-based multiclass approach delivers comparable rank-ordering performance to standard regression as measured by D_{xy} while achieving better calibration, as evidenced by lower $RMSE$ across both original and log-transformed scales. In addition, we examine the calibration of the resulting estimates with Expected Calibration Error (ECE) on the original LGD distribution.

Figures 2 and 3 show the calibration results: the quantile multiclass approach achieves an ECE of 1.4%, compared to 11% for standard regression.

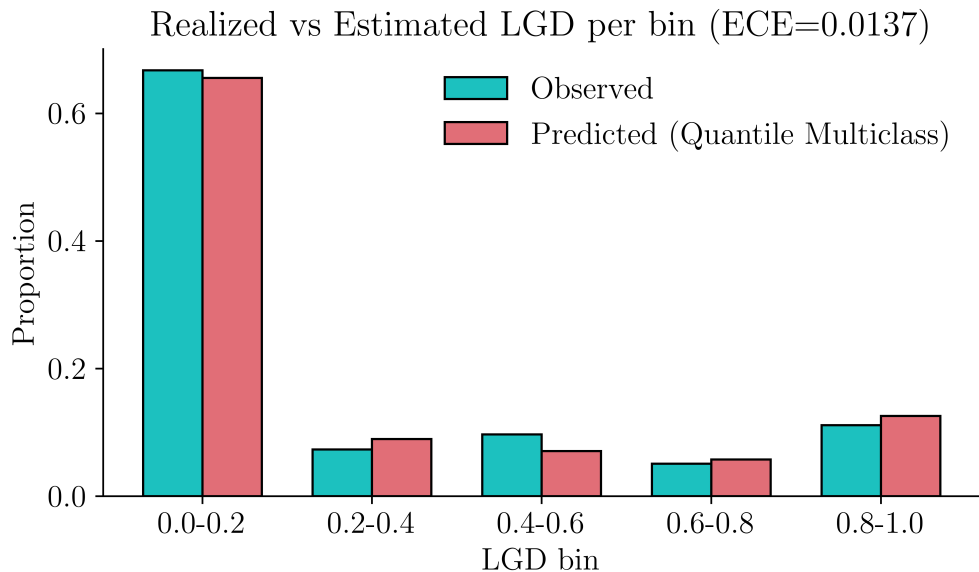


Figure 2: Quantile Multiclass Calibration

The quantile multiclass approach matches standard regression on predictive metrics while achieving 87% better calibration (ECE : 1.4% vs 11%), demonstrating its suitability for applications requiring well-calibrated probability distributions.

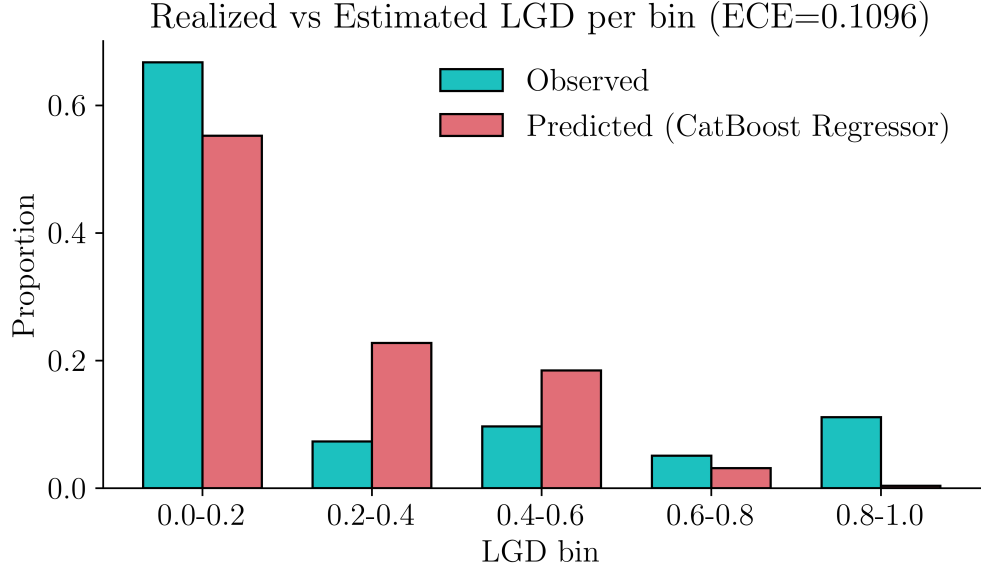


Figure 3: Regression Calibration

3 Conclusion

This paper presented a quantile-based multiclass classification approach for LGD modeling that addresses the inherent bimodality and skewness of LGD distributions.

Our experimental results demonstrate that this framework achieves competitive predictive performance compared to standard regression while providing substantially superior calibration and native support for probability distributions.

The key advantage of this approach lies in its ability to serve multiple regulatory and business requirements from a single model: computing downturn LGD percentiles for capital adequacy, performing stress testing through distribution shifting, and enabling scenario-based expected credit loss calculations under IFRS 9.

By reformulating LGD estimation as classification over quantile bins, the method provides the interpretability and efficiency of traditional approaches while addressing their fundamental limitation, the inability to capture distributional uncertainty.

Future work will explore LGD prediction intervals and Risk Weighted Assets (RWA) simulations using the proposed methodology.

A Appendix

All code and data used in this study are available at: <https://github.com/deburky/lgd-scoring-models/>.

References

- [1] Baesens, Bart and Roesch, Daniel and Scheule, Harald (2016). *Credit Risk Analytics: Measurement Techniques, Applications, and Examples in SAS*. John Wiley & Sons.
- [2] Engelmann, Bernd and Rauhmeier, Robert (2011). *The Basel II Risk Parameters: Estimation, Validation, Stress Testing—with Applications to Loan Risk Management*. Springer Science & Business Media.
- [3] Grisel, Olivier (2025). Probabilistic regression models: let's compare different modeling strategies and discuss how to evaluate them. Talk at PyData Paris 2025, October 1. Slides: https://docs.google.com/presentation/d/1FwMMHPGWzMnEb3eoDpDd_XR7wik-MJUti0rSj6C3ofE/edit?usp=sharing. Code: https://github.com/ogrisel/notebooks/blob/master/probabilistic_regression_figures.ipynb.